LUIGI BERDUCCI

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Senior AI Engineer - Reinforcement Learning at RIVR, an Amazon company

Engineer specialized in **robot learning** and **safe autonomy**: reinforcement learning and sim-to-real transfer.
PhD research in safe RL (TU Wien); industry experience developing RL policies for wheeled-legged robots.

EMPLOYMENT

RIVR | **Senior AI Engineer** - AI Team 02/2025 - Present

- Drive initiatives for **physical intelligence**, focusing on mobility and autonomy of RIVR robots.
- Developed **RL policies** for van climbing, package dropping and navigation in unseen environments.
- Contributed to recruiting, onboarding and **mentoring** of junior engineers.

  Tools: Isaac Sim, Isaac Lab, PyTorch, ROS 2, sim-to-real transfer

Swiss-Mile Robotics | **Robotics Intern** - AI Team 07/2024 - 10/2024

- Student-teacher RL for **locomotion** in industrial use cases with wheeled-legged robots (RIVR One).
- Improved the reinforcement learning pipeline to leverage **symmetries** in robot and environment.
- Refined **domain randomization** models for better sim-to-real transfer.

  Tools: Isaac Sim, Isaac Lab, PyTorch, ROS 2, sim-to-real transfer

University of Pennsylvania | **Visiting Associate** - xLab (GRASP) 02/2023 - 05/2023

- Reinforcement learning to **adapt safety levels** for long-term performance in multi-agent systems.
- Designed a **safety filter** (Control Barrier Function) for collision avoidance in autonomous racing.
- Supported development and maintenance of the F1Tenth racecar **simulator**.

  Tools: Python, PyTorch, CleanRL, OpenAI Gym, CasADi, CVXPY, vehicle-dynamics simulation

TU Wien | **University Assistant** - CPS Group 10/2020 - 12/2024

- Research on **safe reinforcement learning** for autonomous robotic systems.
- Developed a **learner-verifier framework** for provably safe model-based RL with adaptive CBFs.
- Developed **racecar_gym**, a multi-agent racing environment built on the Bullet physics simulator.
- **Supervised** theses and projects on reinforcement learning, deep learning and computer vision.

  Tools: Python, PyTorch, TensorFlow, Stable-Baselines3, MBRL-Lib, CasADi, PyBullet, CARLA

PROJECTS

F1Tenth Autonomous Racing Team (Scuderia Segfault), TU Wien | **Team Member** 09/2021 - 06/2024

- **Grand Prix results: 1st** (IROS 2021), **3rd** of 38 (ICRA 2022), **2nd** of 41 (ICRA 2023), **4th** of 20 (ICRA 2024).
- Developed **planning and control** algorithms for **LiDAR-based navigation** in ROS 2, Python and C++.
- Deployed software on **embedded devices** (NVIDIA Jetson) for F1Tenth vehicles.
- Implemented RL, RRT and Frenet **planners** with domain randomization for real-world racing.

  Tools: C++, Python, ROS 2, Stable-Baselines3, PyTorch, TorchScript, Linux, Git, Docker

EDUCATION

TU Wien, Vienna | **PhD Student** 10/2020 - 12/2024

Thesis: *A Framework for Safety in Reinforcement Learning*

Sapienza University of Rome, Rome | **BSc and MSc in Computer Science** 01/2014 - 05/2020

Full marks with honours (MSc avg. grade 30.3/31.0, BSc avg. grade 27.7/31.0)

SKILLS

Programming: Python, C++
Machine Learning: PyTorch, TensorFlow, scikit-learn
Reinforcement Learning: Isaac Lab, Stable-Baselines3, OpenAI Gym, CleanRL, MBRL-Lib
Robotics and Simulation: ROS 2, Isaac Sim, PyBullet, CARLA, sim-to-real transfer
Optimization: CasADi, CVXPY, Gurobi
Tools: Docker, Git, Linux